Shizhen Jia

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Objective

Ph.D. candidate in wireless communications at NYU WIRELESS seeking internship opportunities in wireless systems and signal processing. Experienced in MIMO beamforming, interference nulling, ray tracing, game-theoretic optimization, FR3 multi-band channel estimation, and OTA phase calibration, with interests in link-level modeling, FR3 spectrum studies, and scalable terrestrial/non-terrestrial communication systems.

Education

- **New York University** *Sep. 2024 – Present*
  Ph.D. candidate in Electrical Engineering, Advisor: Sundeep Rangan New York
- **New York University** *Sep. 2022 – Aug. 2024*
  M.S. in Electrical Engineering New York
- **Chongqing University** *Sep. 2018 – Jun. 2022*
  B.Eng. in Electrical Engineering China

Experience

- **CableLabs — NTN Study Intern** *May – Aug. 2026*
  Louisville, Colorado, United States | Tools: MATLAB, Python
  - Reproduced 3GPP TR 38.863 NTN–TN coexistence results and extended the simulations to multi-satellite systems; developed tools for LEO spatial reuse and multi-satellite NTN–NTN interference evaluation.
  - Developed satellite scheduling algorithms and evaluated capacity, interference, and throughput-loss trade-offs to support spectrum-sharing studies.
  - Built Starlink telemetry and measurement dashboards using gRPC, latency/throughput tests, route diagnostics, and TLE-based serving-satellite estimates; tested RadioHound hardware and investigated Wi-Fi antenna orientation and unlicensed-band NTN operation.

Research Projects

- **Intra-Cell Spatial Reuse in LEO Downlinks: Beam-Leakage-Aware Satellite Scheduling** *2026*
  NYU WIRELESS / CableLabs | Tools: MATLAB, Python, satellite ephemerides
  - Modeled concurrent Ku-band downlinks with realistic satellite geometry, receive-array leakage, antenna patterns, and atmospheric losses; evaluated a 24-hour Starlink trace with 720 snapshots.
  - Developed beam-leakage-aware branch-and-bound scheduling with an admissible upper bound and greedy-plus-swap initialization, certifying optimality for the reference-point model.
  - Improved simulated mean per-beam capacity by 27.2% over nearest-satellite selection with eight beams; reduced mean search time from 529.48 to 1.98 ms when selecting eight of 28 candidates.

- **NTN–NTN Downlink Adjacent-Channel Coexistence** *May – Aug. 2026*

  CableLabs | Tools: MATLAB, Python | Reference: 3GPP Feature Lead and Way Forward since Dec. 2025

  - Built a 2 GHz multi-satellite LEO–LEO / LEO–GEO downlink interference simulator with seven-beam satellite patterns, configurable geometry, 3GPP propagation, and ITU-R P.618 atmospheric loss.
  - Swept satellite geometry and adjacent-channel interference ratio (ACIR) to quantify aggregate interference from multiple satellites and the resulting SINR and throughput degradation, using a 5% loss criterion.
- **NTN–TN Coexistence Across Six Interference Scenarios** *May – Aug. 2026*

  CableLabs | Tools: MATLAB, Python | Reference: 3GPP TR 38.863

  - Reproduced 3GPP TR 38.863 coexistence results and extended the framework to multi-satellite systems with aggregate interference.
  - Implemented six satellite–terrestrial DL, UL, and cross-link scenarios with urban/rural deployments, 19-site/57-sector wrap-around networks, and multi-beam LEO/GEO configurations.
  - Integrated transmitter leakage, receiver selectivity, uplink resource allocation, and power control; used Monte Carlo and ACIR sweeps to evaluate mean and fifth-percentile throughput loss against a 5% criterion.
- **Noncoherent TN–NTN Coexistence: Blind Sensing and Interference Nulling** *Jan. 2026 – Present*

  Advisor: Sundeep Rangan | Tools: Python, Sionna-RT Paper | Project

  - Derive a covariance-eigenvector GLRT and large-matrix/finite-array leakage scaling, linking sensing SNR and snapshot count to residual INR.
  - Develop blind MUSIC sensing, anonymous power estimation, and cross-carrier direction transfer for fixed-power null steering without instantaneous channel reciprocity.
  - Build 10×10 km, 12-sector ray-tracing evaluations: at 8 GHz with analytic covariance and $\lambda = 10^{11}$, increase NTN protection (INR ≤ -3 dB) from 65.3% to 85.2%, with 1.05 dB TN median SINR loss.
  - Evaluate finite-snapshot detection, intermittent uplink activity, and listening/switching overhead; distinguish activity coverage from final-beam protection.
- **Coherent TN–NTN Coexistence: Beacon-Aided Estimation and Interference Nulling** *Sep. 2024 – Jun. 2025*

  Advisor: Sundeep Rangan | Tools: Python, Sionna-RT Paper | Poster

  - Design beacon-based detection, reciprocal-channel estimation, and regularized null steering via a principal-eigenvector beamformer.
  - Derive residual-INR moments, an all-SNR bound, and finite-array low-/high-SNR limits under channel-estimation error; distinguish the high-SNR plateau from an interference ceiling.
  - Build a 15×15 km CableLabs-area ray-tracing study with 7 sites, 21 sectors, 8×8 arrays, and paired true/noisy-CSI evaluations over 10 deployments.
  - With noisy CSI and $\lambda = 10^{13}$, reduce median NTN INR from -12.7 to -27.7 dB while quantifying TN cost: median SINR $20.2 \rightarrow 17.0$ dB and tenth percentile $10.2 \rightarrow -3.4$ dB.
- **Distributed LEO Uplink Anti-Jamming via Game-Theoretic Beamforming** *May 2025 – Nov. 2025*

  Advisor: Sundeep Rangan | Tools: Python, Sionna-RT Paper | Poster

  - Formulate a convex–concave transmit-covariance game; implement water-filling best responses and projected-gradient jammer updates.
  - Evaluate 1/3/5-satellite cooperation over 400 Starlink time frames with Sionna RT, comparing fixed dish and adaptive array jamming under an equalized-channel model.
  - At 1 km dish-jammer separation, three satellites exceed 7.5 bit/s/Hz in over half the simulated cases; quantify achievable-rate gains versus jammer proximity.

- **Delay–Doppler Synchronization for Distributed LEO Interference Nulling** *Ongoing*
  Advisor: Sundeep Rangan | *Tools: Python, Sionna-RT* Paper | Project
  - Co-develop two-branch delay–Doppler alignment and residual-INR analysis, separating acquisition errors, local mismatch, and finite-training weight-learning losses.
  - Reduce measured CPU search/refinement time from 99.0 to 14.7 ms (6.7×) with coarse FFT/gradient acquisition, at a 0.7-dB threshold penalty versus dense FFT/zoom.
  - Validate on independent test blocks and 716 Starlink snapshots; achieve -21.94 dB median snapshot-mean residual INR in the interference-only study.
- **Narrowband OTA Phase Calibration: Multipath Statistics & Spatial Averaging** *Aug. 2025 – Present*
  Advisor: Sundeep Rangan | *Collaborator: Andrea Bedin* | *Tools: Pi-Radio Marie Curie, RFSoC, MATLAB, Python* Paper | Poster
  - Characterize narrowband phase errors across 26 indoor runs at five carriers (8.75–14.9 GHz), using coherent Pi-Radio/RFSoC measurements over 300 transmitter positions.
  - Derive phase-error statistics under a dominant-path/weak-residual model; obtain smaller empirical CDF discrepancies than a Gaussian baseline in all 30 pooled bandwidth–frequency settings.
  - Develop circular averaging of normalized two-channel ratios; quantify spatial correlation, residual bias, and sampling reliability against a paired wideband reference.
  - At 5% bandwidth and 598 mm coverage, systematic spatial sampling reduces median RMSE from 3.36° to 2.37° (29.5%) with 50 observations, improving 128/130 configurations.
- **DMC Multi-band Channel Estimation in FR3 Band** *Dec. 2025 – Present*
  Advisor: Sundeep Rangan | *Tools: Pi-Radio Marie Curie, RFSoC, MATLAB, Python*
  - Implement Newtonized OMP (NOMP) for FR3 multi-band DMC channel estimation (delay/angle refinement).
  - Prepare OTA multi-band experiments on the Pi-Radio/RFSoC platform to validate DMC models.
- **Interference Avoidance for FMCW Automotive Radars via Non-Cooperative Game** *Aug. 2023 – Aug. 2024*
  M.S. Thesis | *Advisor: Quanyan Zhu* | *Tools: MATLAB, Python*
  - Developed a system-level interference framework for FMCW automotive radar networks.
  - Built an analytical collision model linking waveform timing to SINR degradation.
  - Formulated spectrum coordination as a non-cooperative game and derived mixed-strategy equilibria.
  - Quantified robustness gains of game-theoretic policies over heuristic baselines via simulation.

Publications

[C.1] **Shizhen Jia**, Mingjun Ying, Marco Mezzavilla, Doru Calin, Theodore S. Rappaport, Sundeep Rangan (2025). **Joint Detection, Channel Estimation and Interference Nulling for Terrestrial–Satellite Downlink Co-Existence in the Upper Mid-Band**. *Proc. IEEE GLOBECOM*, accepted, Dec. 2025.

[C.2] **Shizhen Jia**, Mingjun Ying, Marco Mezzavilla, Theodore S. Rappaport, Sundeep Rangan (2026). **Distributed Uplink Anti-Jamming in LEO Mega-Constellations via Game-Theoretic Beamforming**. *Proc. IEEE ICC*, accepted, 2026.

[C.3] **Shizhen Jia**, C. Nicolas Barati, Marco Mezzavilla, Sundeep Rangan (2026). **Noncoherent Detection and Interference Nulling for Terrestrial–Satellite Downlink Coexistence in the Upper Mid-Band**. *Proc. IEEE SPAWC*, 2026.

Professional Activities

- **Session Chair** *Dec. 2025*
  IEEE GLOBECOM 2025
  - Channel Modeling & Estimation (SAC-SSC-03), December 10, 2025.
- **Journal Reviewer**
  - IEEE Communications Letters
  - IEEE Transactions on Wireless Communications (TWC)
  - IEEE Transactions on Cognitive Communications and Networking (TCCN)
- **Conference Reviewer**
  - IEEE Global Communications Conference (GLOBECOM)
  - IEEE International Conference on Communications (ICC)